

TOGAF® ENTERPRISE ARCHITECTURE

M015 TOGAF Overview

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BASIC CONCEPTS

What is an Enterprise

What is Enterprise Architecture

What is an EA Framework

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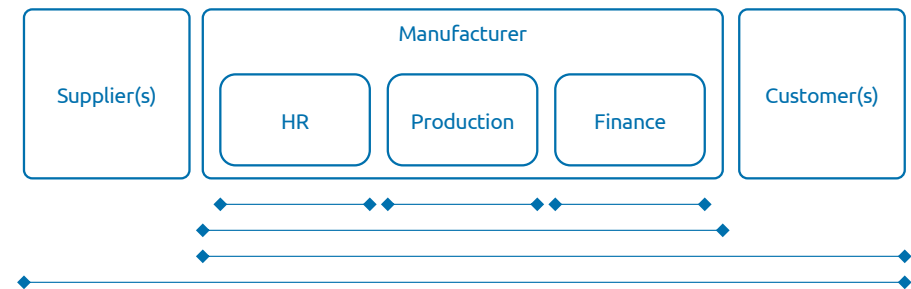




TOGAF OVERVIEW

01 DEFINITION OF AN ENTERPRISE

- An **enterprise** is considered to be a collection of organisations that have **common goals**
- The term Enterprise can apply to:
 - an **entire enterprise** and all of its business activities and capabilities, information, and technology
 - or **one or more specific** areas of interest within the enterprise
- An enterprise may **include** partners, suppliers, and customers as well as internal business units
- The architecture crosses multiple **systems**, and multiple **functional groups** within the enterprise
- Organisations may comprise **multiple enterprises**, and develop and maintain a number of **independent** Enterprise Architectures to address each one



Examples include:

- a whole corporation or a division of a corporation
- a government agency or a single government department
- a chain of geographically distant organisations linked together by common ownership
- groups of counties, governments, or governmental organisations (such a militaries) working together to create common or shareable deliverables or infrastructures



TOGAF OVERVIEW

01 PURPOSE OF ENTERPRISE ARCHITECTURE

Purpose of an Enterprise Architecture

- An Enterprise Architecture is developed **to guide effective change**
- Enabling the **right balance** between business transformation and continuous operational efficiency
- Helping to **optimise** fragmented legacy processes into an integrated environment that is **responsive to change**
- Supporting **effective management and exploitation** of information, and digital transformations by providing strategic context in a constantly changing business environment
- Providing a framework for change, linked to both **strategic direction** and **business value**

Importance of Enterprise Architecture

- An architected approach provides a rigorous **planning** and change **governance** methodology
- Enterprise Architecture must describe the **future state** and **current state** of the enterprise:
 - future state description enables the right stakeholders to **understand what must be done** to meet goals, objectives, mission and vision of the enterprise
 - gaps between current and future states **highlights what must be change**
 - knowing what gaps exist makes it obvious what must change and the implications of that change
- A good enterprise architecture facilitates **effective** governance, management, risk management, and exploitation opportunities
- EA Governance can be considered as two parts:
 - To **direct** change activity
 - To **control** change activity

There is **no** single correct scope, level of detail, or purpose for EA – different enterprises will expect their EA to guide change at different levels in the enterprise



TOGAF OVERVIEW

01 VALUE OF ENTERPRISE ARCHITECTURE

The range, scope, and scale of an Enterprise Architecture are as **broad** as the scope of and scale of **enterprises** and their change programmes. The ability to develop, use, and sustain the required Enterprise Architecture will be equally as broad.

However, the **value** of **Enterprise Architecture** is consistent:

More effective strategic decision making by C-level executives and business leaders	More effective and efficient business operations	More effective and efficient Digital Transformation and operations	Better return on existing investment, reduced risks for future investment	Faster, simpler, and cheaper procurement
▪ Quicker response to change, aligned with strategy ▪ Support for enterprise agility ▪ Organisational transformation, adopting new business and technology trends ▪ Organisational and operating model changes to improve efficiency and effectiveness	▪ Lower costs (business operation, change management) ▪ More agile organisation ▪ Business capabilities shared across the organisation ▪ More flexible workforce ▪ Improved business productivity ▪ Improved organisation integration in support of mergers and acquisitions	▪ Extending reach of the enterprise (e.g. through digital capability) ▪ Bringing all components into a harmonised environment ▪ Lower development, deployment, operations, support, and maintenance costs ▪ Improved interoperability ▪ Improved system management ▪ Easier upgrade and exchange of system components	▪ Maximise return on investments in existing business and IT ▪ Flexibility to make, buy, or outsource business and IT solutions ▪ Understanding how investment changes over time	▪ Simpler buying decisions ▪ Information governing procurement is readily available ▪ Faster procurement ▪ Ability to procure heterogeneous, multi-vendor open systems ▪ Ability to secure more economic capabilities

Enterprise Architecture provides a strategic context for the evolution of the enterprise to the constantly changing needs of the business environment



TOGAF OVERVIEW

01 ENTERPRISE ARCHITECTURE AND FRAMEWORKS

Enterprise Architecture is:

- A description of the **elements** within an organisation:
 - what they are meant to **achieve**
 - how they are **arranged**
 - how they **perform** in practice
 - how they **respond to change**
- A framework (structure, approach, process) for **managing change** to those elements and their arrangement; to continually adapt to organisational change in line with strategy and circumstances
- The **practice** of acting to manage and evolve the Enterprise Architecture at all levels of control, change, and pace

An Enterprise Architecture Framework is:

- A **foundational** structure, or set of structures, which can be used for developing a **broad range** of different architectures
- With a method for describing both a **baseline** and a **target** state of the enterprise
- As a set of **building blocks** for showing how the building blocks fit together
- Typically **tailored** to meet the specific needs of an organisation
- **Tailored** to establish a set of tools and a common vocabulary

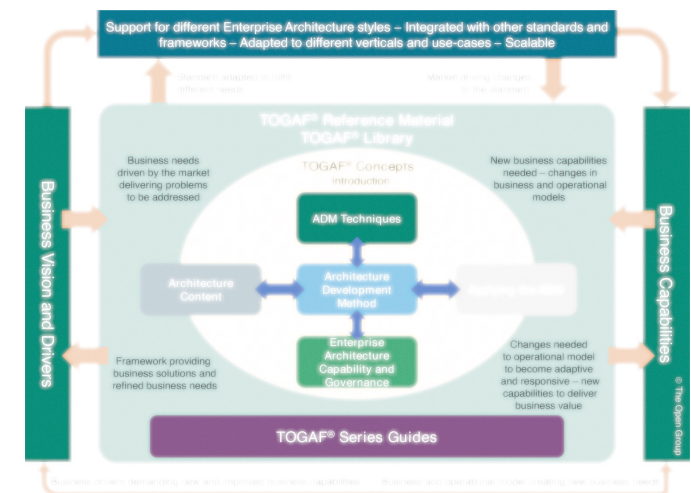


TOGAF OVERVIEW

01 TOGAF AS AN ARCHITECTURE FRAMEWORK

The TOGAF Standard is:

- A framework for **identifying** and **implementing** change, providing:
 - A definition and description of a **standard cycle** of change
 - Used to plan, develop, implement, govern, change, and sustain an architecture for an enterprise
 - A definition and description of the **building blocks** in an enterprise used to deliver business services and information systems
 - A set of guidelines, techniques, and advice to create and maintain an effective Enterprise Architecture
 - Deliver change through new Solution Architectures at all levels of scale, pace, and detail
- Results in **Enterprise Architecture** that is **consistent**, reflects the needs of **stakeholders**, employs **best practice**, and gives due consideration to both **current** requirements and perceived **future** business needs
- Plays an important role in **standardising** and **de-risking** the architecture development process
- Provides a best practice framework for **adding value**, and enables an organisation to build **workable** and **economic** solutions that address business needs



TOGAF is an architecture framework — The Open Group Architecture Framework

TOGAF INTRODUCTION

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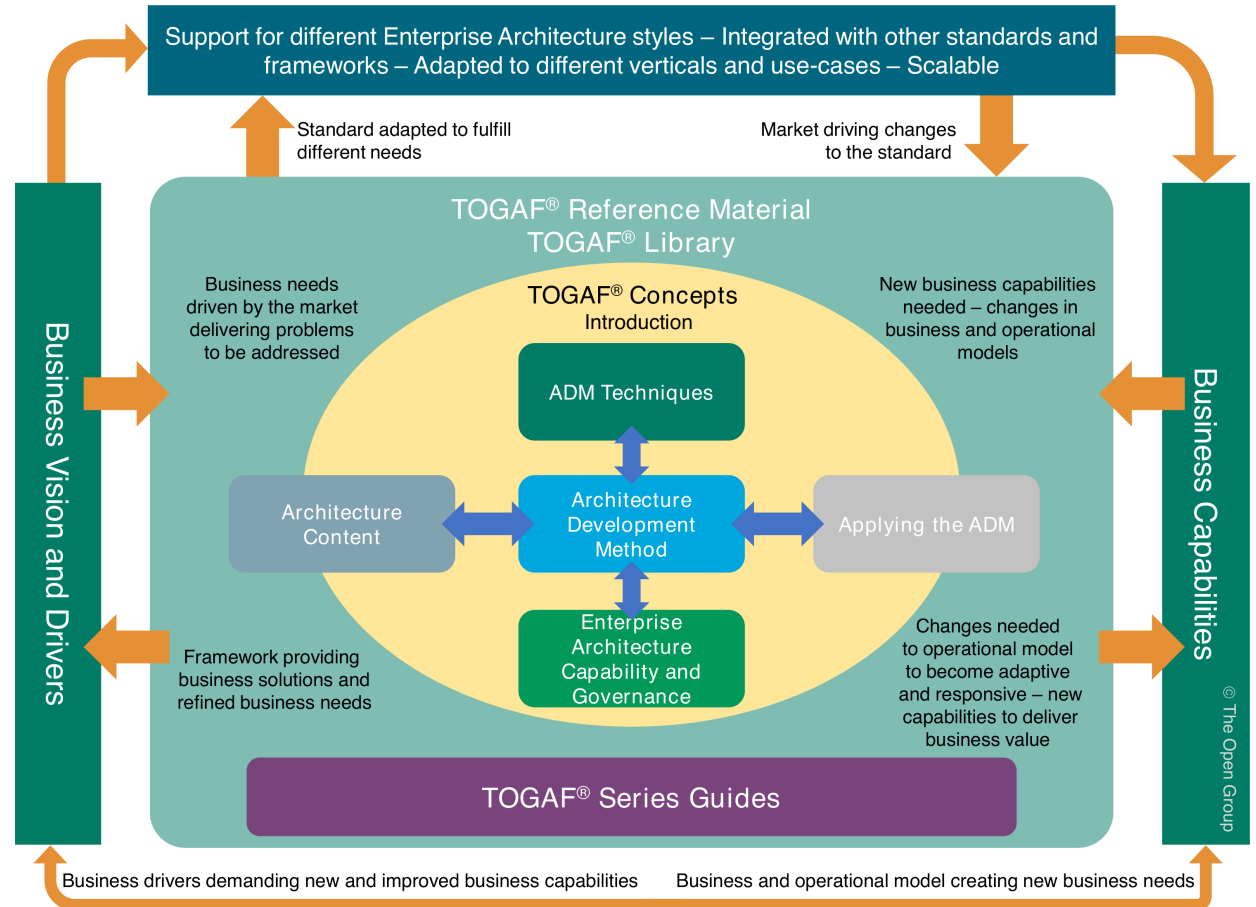




TOGAF OVERVIEW

02 TOGAF INTRODUCTION

- The TOGAF Standard is an **open**, industry **consensus** framework for Enterprise Architecture
- It is a **foundational framework** which is applicable to the development of any kind of architecture in any context
- This foundational framework is supplemented by The Open Group **TOGAF Library**, an extensive and growing portfolio of **guidance material**
- The structure of the TOGAF Standard **reflects** the structure and content of an **Architecture Capability** within an enterprise

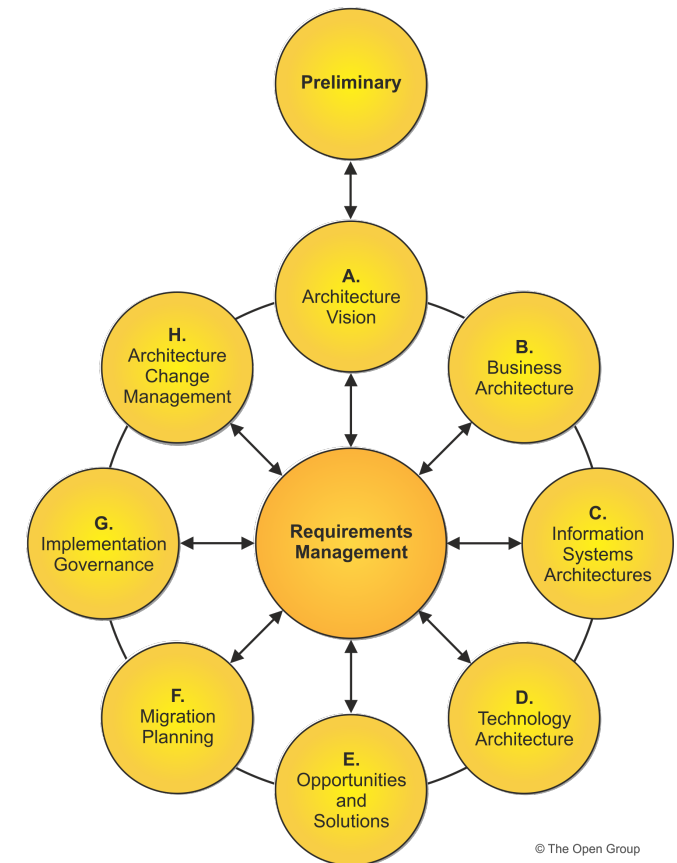




TOGAF OVERVIEW

02 ARCHITECTURE DEVELOPMENT METHOD (ADM)

- The ADM describes a method for developing and managing the **lifecycle** of an **Enterprise Architecture**
- The ADM forms the core of the TOGAF Standard, and includes
 - establishing an **architecture framework**
 - developing **architecture content**
 - transitioning and governing the **realisation of architectures**
- It is **iterative**, over the whole process, between phases, and within phases
- It is **adaptive** and can be **tailored** to suit the organisation
- It **does not** mandate a “waterfall” method, nor does it mandate that all phases must be performed in any specific sequence
- It **integrates elements** of the TOGAF Standard, as well as other available architectural assets, to meet the business needs of an organisation



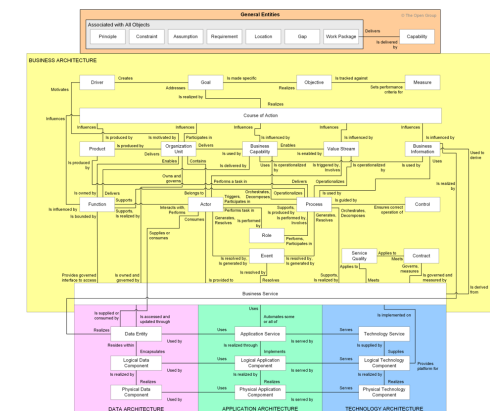
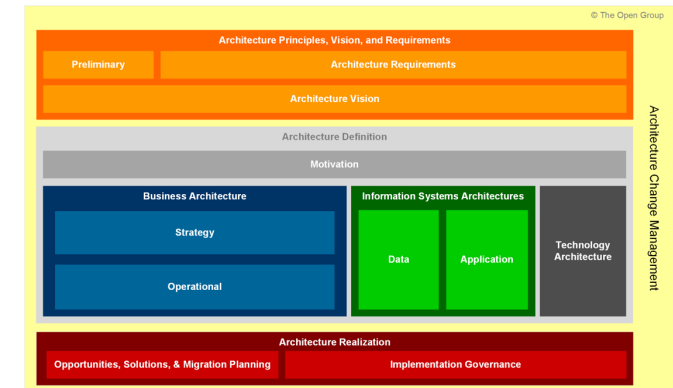
The TOGAF Architecture Development Method (ADM) provides a tested and repeatable process for developing architectures



TOGAF OVERVIEW

02 CONTENT FRAMEWORK AND ENTERPRISE METAMODEL

- The **Content Framework** is a categorisation framework to describe the building blocks and artifacts reflecting decisions taken in creating architecture deliverables
- An **Enterprise Metamodel** defines the entity types and their relationships that describe the enterprise in artefacts and models
- There are **four architecture domains** supported by TOGAF:
 - **Business Architecture** defines business strategy, governance, and key business processes
 - **Data Architecture** describes the structure of an organisation's logical and physical data assets, and data management resources
 - **Application Architecture** provides a blueprint for the individual applications to be deployed, their interactions, and their relationships to the core business
 - **Technology Architecture** describes the digital architecture and logical software and hardware infrastructure and standards to support the deployment of business, data and application services



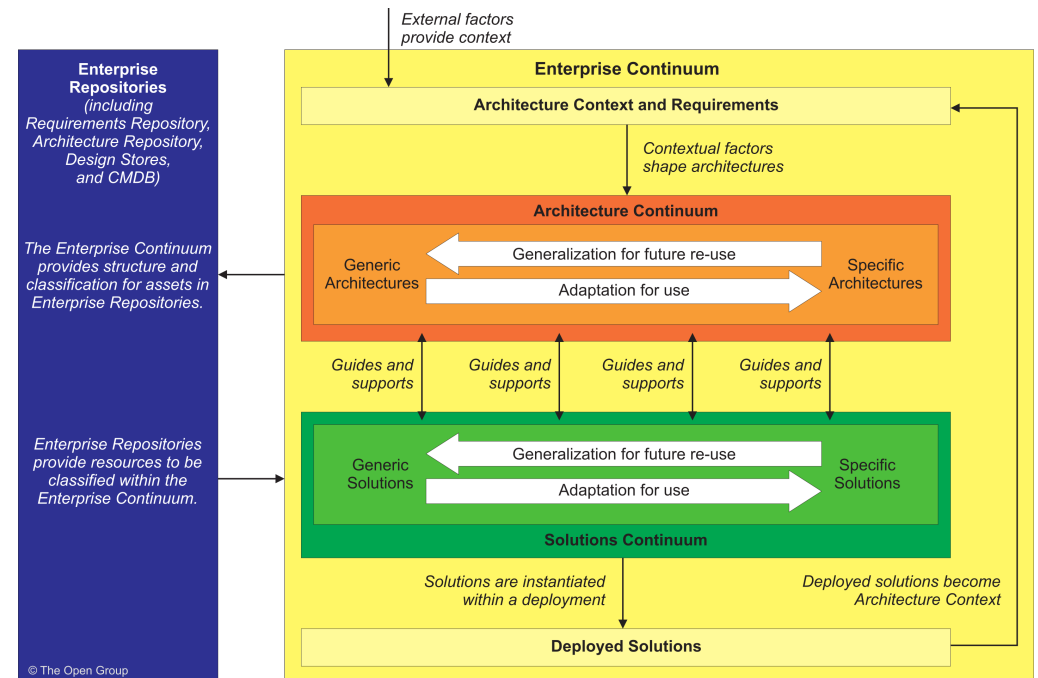
The ADM, Content Framework and Foundational Enterprise Metamodel can be utilised together, but the TOGAF Standard can be adapted to use other content frameworks and enterprise metamodels from other architecture framework and methods.



TOGAF OVERVIEW

02 ENTERPRISE CONTINUUM

- The **Enterprise Continuum** is a **categorisation** for assets held in the **Enterprise Repositories**
- Providing methods for classifying assets as architecture and solution artifacts evolve from generic Foundation Architectures to Organisation-Specific Architectures
- The continuum can classify:
 - architecture descriptions
 - models
 - building blocks
 - Patterns
 - architecture viewpoints, and other artifacts
 - that exist both within the enterprise and in the IT industry at large
- The **Enterprise Continuum** is partitioned into **two distinct continua** as follows:
 - Architecture Continuum
 - Solutions Continuum



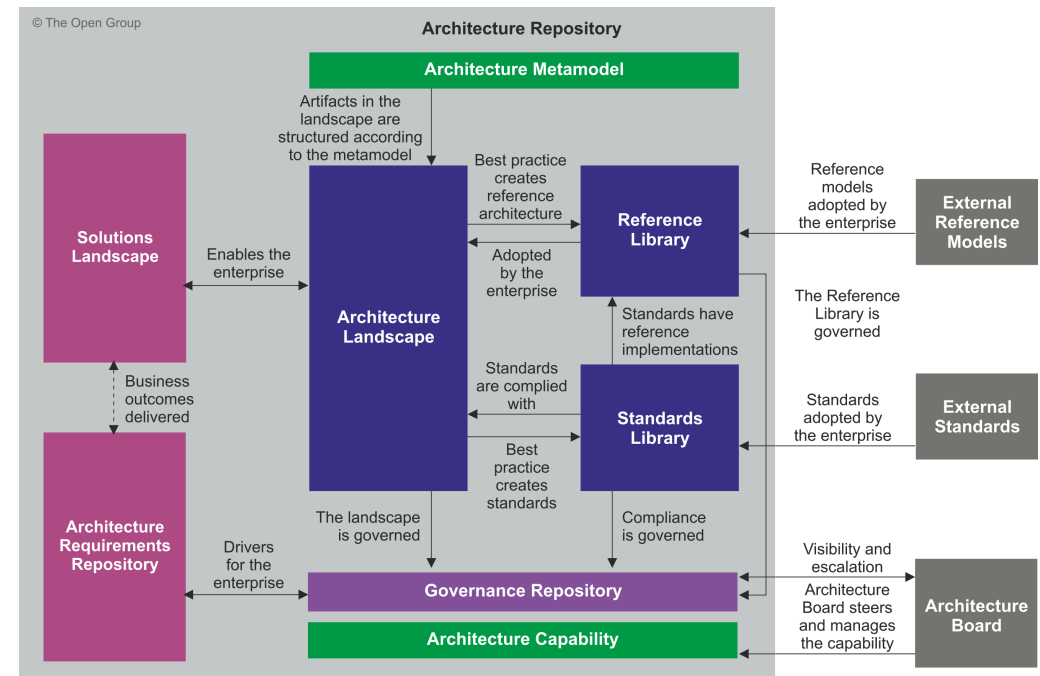
The simplest way of thinking of the Enterprise Continuum is as a view of the repository of all the architecture assets



TOGAF OVERVIEW

02 ARCHITECTURE REPOSITORY

- Supporting the **Enterprise Continuum** is the concept of an **Architecture Repository**, to store:
 - different classes of architectural output
 - at different levels of abstraction
 - created during execution of the **ADM**
- The following classes of architectural information are expected to be held within an architecture repository:
 - Architecture Metamodel
 - Architecture Capability
 - Architecture Landscape
 - Standards Library
 - Reference Library
 - Governance Repository
 - Architecture Requirements Repository
 - Solutions Landscape



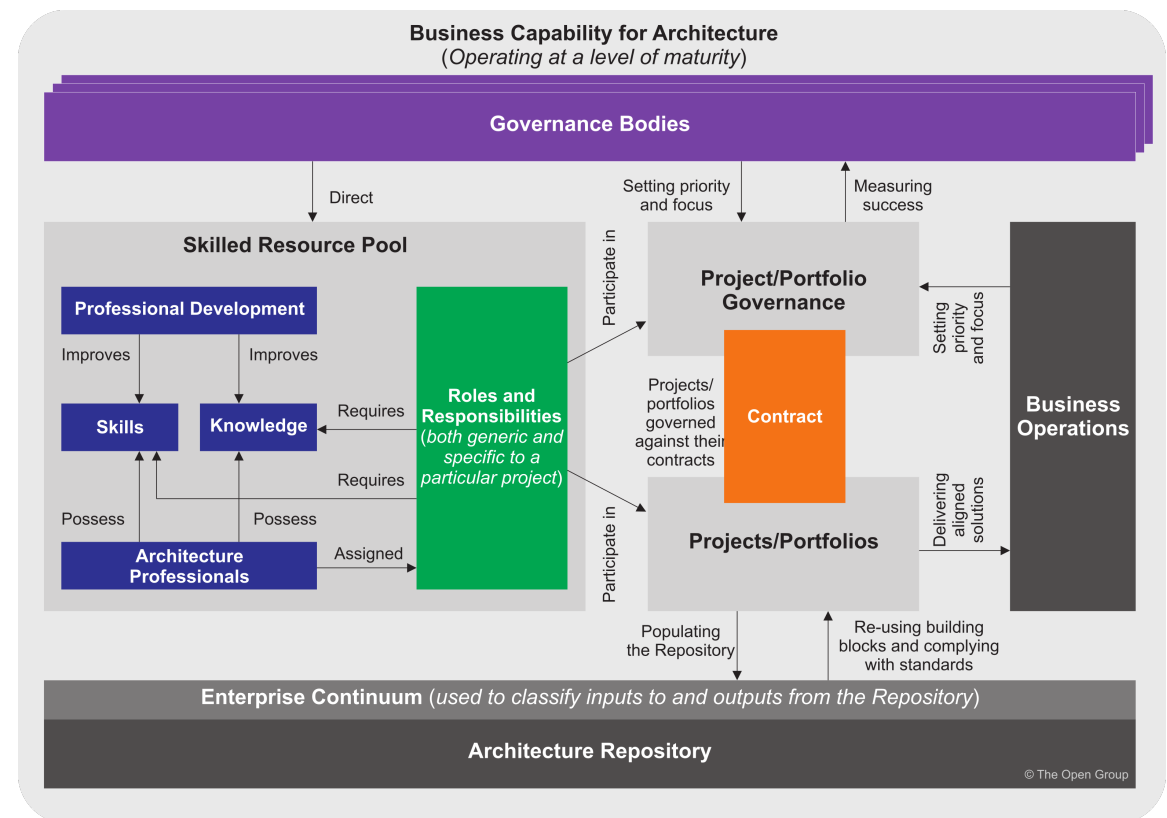
The TOGAF ADM can be regarded as describing a process lifecycle that operates at multiple levels within the organisation, operating within a holistic governance framework and producing aligned outputs that reside in an Architecture Repository



TOGAF OVERVIEW

02 ESTABLISHING AN ARCHITECTURE CAPABILITY

- An appropriate **business capability for architecture** is necessary to undertake architectural activities effectively
- This involves organisation structures, roles, responsibilities, skills, and processes
- An **Enterprise Architecture Practice** must be **run like any other** operational unit within a business
- Well-defined and effective **governance** is central to on-going architecture practice
- Where all architecturally significant activity is controlled and aligned within a single framework



It is increasingly recognised that a successful Enterprise Architecture Practice must sit on a **firm operational footing**

THE TOGAF LIBRARY

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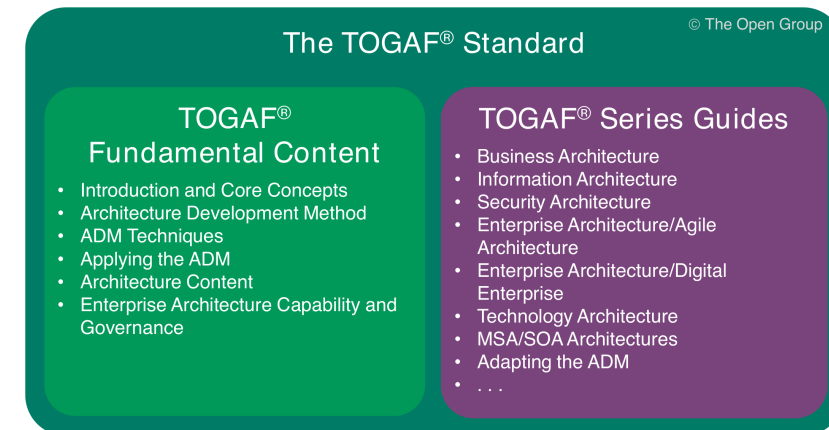




TOGAF OVERVIEW

03 THE TOGAF STANDARD

- The TOGAF Standard consists of:
 - **TOGAF Fundamental Content** core content in six free-standing documents;
 - **TOGAF Series Guides** providing guidance that address specific concerns and use-cases
- The TOGAF Series Guides support more **specific** needs from practitioners who need further explanation, or more **detail** than is provided in the core content
- Structuring the TOGAF Standard this way allows:
 - for different areas of **specialisation** to be considered in detail, and potentially addressed in isolation
 - **selection** of particular documents for **adoption** while excluding others
- Allowing the Standard to be **evolved more easily**
- Enabling organisations to incrementally adopt elements at **their own pace and cadence**
- Allowing guidance to **evolve** more **rapidly** than core content to cover new needs as they emerge from market trends and the need of industry



Examples of areas covered by guidance material are:

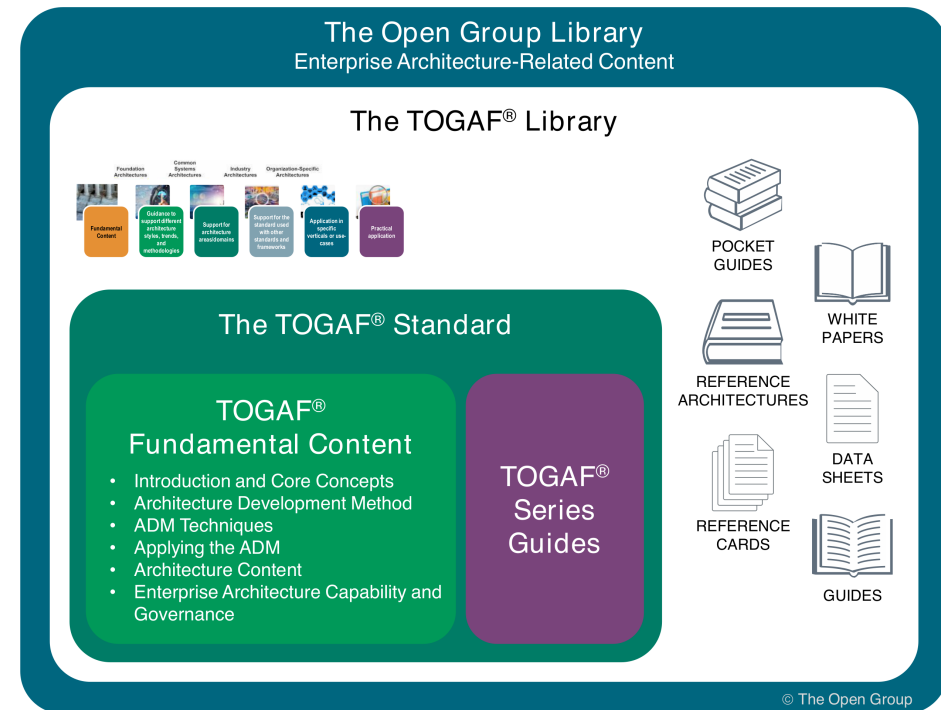
- Strategy decision-making and business value-oriented decisions
- Business Architecture and operating model description
- Information and data management
- Information reference models and data integration models
- Technology Architecture and how EA can be used to adopt new technology trends
- Security Architecture and how TOGAF can be applied as a standard to deliver and support Security Architecture
- How EA and TOGAF can be applied to support the agile enterprise



TOGAF OVERVIEW

03 THE TOGAF LIBRARY

- Accompanying the TOGAF Standard is a broad portfolio of guidance material named the **TOGAF Library**
- To support the practical application of the TOGAF approach
- The **TOGAF Library** is a **reference library** containing:
 - guidelines
 - templates
 - patterns
 - other forms of reference material
 - to accelerate the creation of new architectures
- The **TOGAF Library** follows a **categorisation model** based on capabilities and features that can be delivered into the market through different sets of documents and resources



LEARNING REVIEW

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TOGAF OVERVIEW

04 LEARNING REVIEW

- An **enterprise** is considered to be a collection of organisations that have **common goals**
- The purpose of **Enterprise Architecture** is **to guide effective change**
- **Enterprise Architecture** must describe the **future state** and **current state** of the enterprise
- The range, scope, and scale of enterprise architecture **is as broad** as the scope of and scale of enterprises and their change programmes
- An Enterprise Architecture Framework is a **foundational structure**, or set of structures, which can be used for developing **a broad range** of different architectures
- The TOGAF Standard is an **open**, industry **consensus** framework for enterprise architecture ; that **reflects** the structure and content of an **Architecture Capability** within an enterprise
- TOGAF comprises the **Architecture Development Method, Content Framework, Enterprise Metamodel, Enterprise Continuum, Architecture Repository**
- The TOGAF Standard consists of **TOGAF Fundamental Content** and **TOGAF Series Guides** to help direct and guide the practitioner
- Accompanying the TOGAF Standard is a broad portfolio of guidance material, known as the **TOGAF Library** to support the practical application of the TOGAF approach

ADDITIONAL READING

- What an Enterprise Architecture Looks Like





TOGAF OVERVIEW

05 ADDITIONAL READING

What an Enterprise Architecture Looks Like...

EA exists to guide and constrain change planning and work to perform the change. The scope of work embedded in a Request for Architecture Work should identify the applicable characteristics of the EA Landscape. Over time, through multiple Architecture Projects, the EA Landscape is populated.

In short, an architect will need to document three things:

- Models, in the EA Landscape
- Views derived from the EA Landscape
- Other useful things

In short, the architecture is the set of models, the components, and their relationships that comprise the scope of the EA Landscape under consideration. These models consistently describe the current and Target Architecture. Models can vary in formality, some strictly conforming to a semantically constrained structure, while others are quite flexible.

The primary purpose of the models is to facilitate the architect to understand the system being examined. Understand how it works today, understand how it can be most effectively changed to reach the aspirations of the stakeholders, and understand the implications and impacts of the change.

A secondary purpose is re-use. It is simply inefficient to re-describe the Enterprise. The efficiency of consistency is balanced against the extra energy to describe more than is needed, and to train those who describe and read the descriptions on formal modelling. The size, geographic distribution, and purpose of the EA team will dramatically impact the level of consistency and formality required. Formal models are substantially more re-usable than informal models. Formal models are substantially easier to extend across work teams. The penalty is that formal models require semantic precision.

For example, regardless of the structure of an application in the real world, it must be represented in a model conforming to the formal definition. This representation is possible with a good model definition.



TOGAF OVERVIEW

05 ADDITIONAL READING

...What an Enterprise Architecture Looks Like

Every model that is produced and maintained has a price in effort. When effort exceeds value, the price will be paid by hindering an Enterprise's ability to perform the effective change. Unnecessary models and analysis steal from guiding effective change. Every approach to modelling is designed to shed light on one or more aspects of the Enterprise. Typically, narrow, special-purpose models facilitate detailed analysis while broad models facilitate inclusive analysis. All approaches to modelling – formal/informal and broad/narrow – are trade-offs.

All EA Landscapes that support a broad range of purposes will be comprised of a set of models. This set could be contiguous or discrete, targeted for analysis or communication. A core unified model can provide a common bridge between discrete models. The more specific a model, the more important it is to an analysis. The more important a model to analysis, the more important is the need and clarity of linkage across models. Careful thought is needed to understand the long-term need for cross-linkage. Most analyses are performed repeatedly over a period of time for different purposes.

Models are very useful for the architect. They form consistent representations of the parts of the world that must be understood and analysed. However, because models are partial representations of the whole, typically described with a limited language that requires experience to read, and often subject to constraints designed to show relationships, models tend to be ineffective to communicate usefully.

Models are poor general communication tools. Good models are carefully constrained to exactly tell part of a story. They will carefully control the components available and the available relationships. They will enforce some attributes. They carefully render a complex environment into something that represents the world in terms it can be understood, optimised, and compared. They tend to require specialist knowledge, and often carefully constrain common terms in a way casual consumers do not align with.

The best communication comes down to views, and “other useful things”. Views have a specialized role in communicating the architecture. The phrase “other useful things” is purposefully open-ended. For example, it is normal to find that a high fraction of useful communication is highlighting the value of the target state, acknowledgment of the scope of anticipated change, or clarifying the date value is expected. Most of the effective communication about an architecture will be “other useful things”.



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